Current Electricity

It deals with electrodynamics (charge in motion).

Whenever charge flows from a body to another, in a given time, it is the current.

Current : Rate of flow of charges.

It is denoted by symbol [I].

Its SI unit is Ampere (A).

$$I = \frac{Q}{t} = \frac{ne}{t}$$

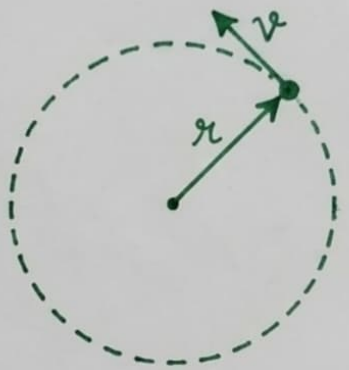
If 'q' charge is a function of time,

$$q = at^2 + bt + c$$

$$I = \frac{dq}{dt}$$

The Revolving Electron & Tensor Nature

If an e^- revolves around a circular orbit of radius 'r' & speed 'v'.



$$v = \frac{2\pi r}{t} \Rightarrow t = \frac{2\pi r}{v}$$

Now, $I = \frac{q}{t} = \frac{ev}{2\pi r}$

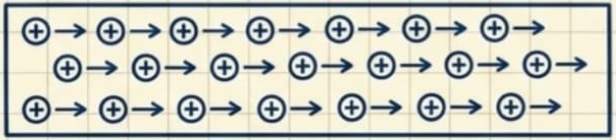
$\therefore I = \frac{e}{T} = e \cdot f$ (where T is time period, f is frequency).

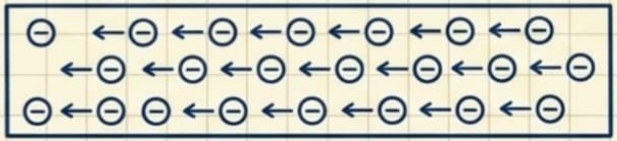
Current as a Tensor Quantity :-

It is a tensor quantity (a special type of scalar quantity). → This means that current has magnitude & direction but is not added by vector addition.

If current is a function of time i.e., $I(t)$, then $Q = \int I\,dt$

Types of Current

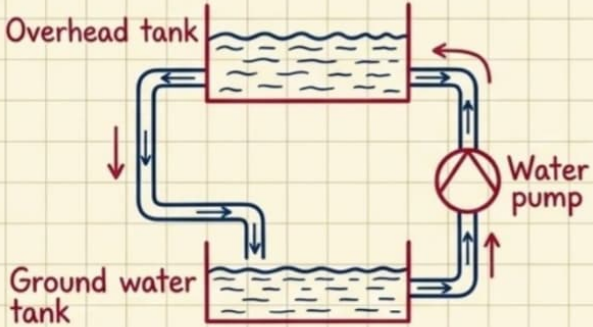


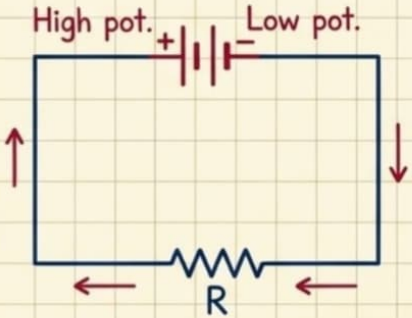


I → Conventional Current:
It is a direction of motion of the +ve charges.

I ← Electronic Current

Maintenance of steady current in a circuit is analogous to the flow of water in a pipe.





Emf vs. Potential Difference (P.d.)

Emf	P.d.
1. It is amt of work done per unit charge to carry it around a complete circuit.	1. It is amt of work done to carry a unit charge b/w any 2 terminals in a circuit.
2. It is denoted by ε.	2. It is denoted by V.
3. Its SI is volt.	3. Its SI is volt.
4. It exists even if circuit is not closed.	4. It only exists when circuit is closed.
5. It is maximum p.d b/w 2 terminals of source.	5. Potential difference can exist b/w any 2 terminals of closed circuit.

When circuit is open, $\varepsilon = V$

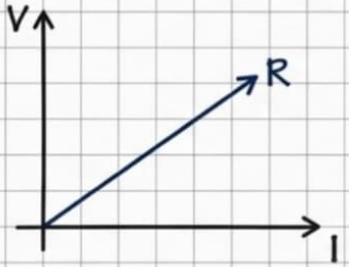
When circuit is closed, $\varepsilon > V$

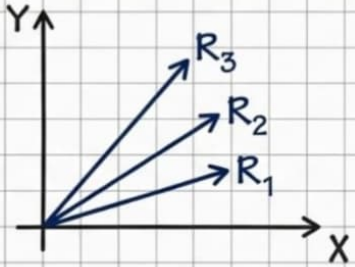
Ohm's Law

1. It is given by George Simon Ohm.
2. It states that current flowing through a circuit is directly proportional to its applied potential difference.
3. At constant temperature & physical conditions.

$$\boxed{\begin{array}{l} V \propto I \\ V = IR \text{ or } R = \frac{V}{I} \end{array}}$$

$\therefore$ Slope of V/I graph = R = tan θ (Slope of graph).

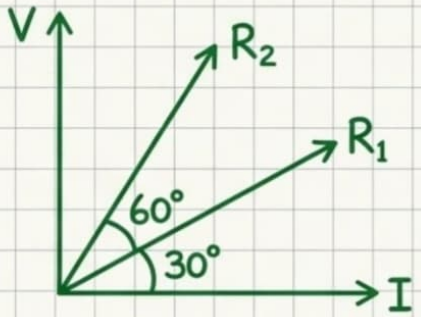




$\therefore R_3 > R_2 > R_1$

SI unit of R is Ω.

Example: Find ratio of $R_1 : R_2$



$$\frac{R_1}{R_2} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{\frac{1}{\sqrt{3}}}{\sqrt{3}} = \frac{1}{3}$$

$$\therefore R_1 : R_2 = 1 : 3$$

Factors affecting resistance of wire / conductor

1. $R \propto \ell$ --- (Equation I)
2. $R \propto \frac{1}{A}$ --- (Equation II)

$\Rightarrow$ Combining I & II, we get: $R \propto \frac{\ell}{A} \Rightarrow R = \rho\left(\frac{\ell}{A}\right)$

Where ρ is resistivity of material used in wire. Its SI is Ω-m.

Conductance & Conductivity

1. Conductance

$$G = \frac{1}{R}$$

This is called as Conductance & its SI is Ω^{-1} or mho.

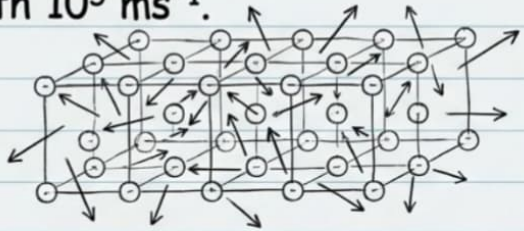
2. Conductivity

$$\sigma = \frac{1}{\rho}$$

This is called as Conductivity & its SI is $\Omega^{-1}m^{-1}$.

Mechanism of Current Flow in Conductor (Absence of Field)

1. Metals have large no. of free e^-, nearly 10^{28} / m^3.
2. In absence of $\vec{E}$ (vector), these e^- move randomly in all directions due to thermal energy.
3. At room temperature, e^- move with 10^5 ms^{-1}.



$\sum \vec{v} = \vec{0}$

Consider that while initially when e^- were moving randomly in all directions, going in a particular direction is equal to the no. of e^- going in opposite direction.

So, if $\vec{u}_1, \vec{u}_2, \vec{u}_3 \ldots \vec{u}_n$ are initial velocities of n e^-.

So, $\vec{u}$ (avg) $= \dfrac{\vec{u}_1 + \vec{u}_2 + \vec{u}_3 \ldots + \vec{u}_n}{n} = 0$ --- (Equation I).

On avg, no. of e^- going in 1 direction equals to no. of e^- going in opposite direction.

Mechanism of Flow: Applying the Field

After applying field $\vec{E}$, each e^- will experience force $[-e\vec{E}]$.

$$\therefore \quad \vec{F} = -e\vec{E}$$

$$\text{Or } m\vec{a} = -e\vec{E} \quad \Rightarrow \quad \vec{a} = \frac{-e\vec{E}}{m} \quad \text{--- (Equation II).}$$

Now, after experiencing this acceleration, the final velocities will change to:

$$\left.\begin{aligned} \vec{v}_1 &= \vec{u}_1 + \vec{a}\cdot\tau_1 \\ \vec{v}_2 &= \vec{u}_2 + \vec{a}\cdot\tau_2 \\ \vec{v}_3 &= \vec{u}_3 + \vec{a}\cdot\tau_3 \\ &\vdots \\ \vec{v}_n &= \vec{u}_n + \vec{a}\cdot\tau_n \end{aligned}\right\}$$

(where τ is the relaxation time b/w 2 successive collisions)

Derivation of Drift Velocity (V_d)

Avg final velocity of e^- drifting through conductor is called as drift velocity (V_d) & is given by:

$$\vec{V}_d = \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 \ldots \vec{v}_n}{n}$$

$$\therefore \vec{V}_d = \frac{[(\vec{u}_1 + a\vec{\tau}_1) + (\vec{u}_2 + a\vec{\tau}_2) + (\vec{u}_3 + a\vec{\tau}_3) \ldots (\vec{u}_n + a\vec{\tau}_n)]}{n}$$

$$\text{Or } \vec{V}_d = \frac{\vec{u}_1 + \vec{u}_2 + \vec{u}_3 \ldots \vec{u}_n}{n} + \frac{\vec{a} \cdot (\vec{\tau}_1 + \vec{\tau}_2 + \vec{\tau}_3 \ldots \vec{\tau}_n)}{n}$$

0

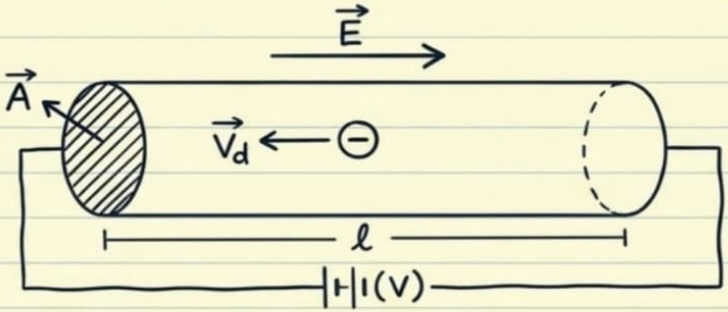
$\therefore \vec{V}_d = \vec{0} + \vec{a} \cdot \vec{\tau}$ (where τ is avg relaxation time).

$$\therefore \vec{V}_d = \vec{a} \cdot \vec{\tau} = \left(\frac{-e\vec{E}}{m}\right) \cdot \vec{\tau}$$

$$\therefore \vec{V}_d = \frac{-e\vec{E}\vec{\tau}}{m}$$

The –ve sign shows that $\vec{V}_d$ is opp to $\vec{E}$ & e^- drift opposite to applied $\vec{E}$.

Current Flow & Current Density



Consider a conductor of length (ℓ), cross-sectional area (A), in which e^- drift in a direction opposite to applied $\vec{E}$.

Here, the applied $\vec{E}$ is given by, $E = \frac{V}{\ell}$.

Consider, there are n e^- per unit volume of this conductor.

No. of $e^- = n \cdot Vol = n \cdot A \cdot \ell$

$\therefore$ charge on e^- (q) $= n \cdot e \cdot A \cdot \ell$

$\therefore I = \frac{n \cdot e \cdot A \cdot \ell}{t}$ Now, $\frac{\ell}{t} = V_d$

$\therefore I = neAV_d$ or $I = V_d enA$.

For current density, J:

$$J = \frac{I}{A} \quad \text{or} \quad J = \frac{V_d enA}{A} = V_d en$$

Microscopic Derivation of Resistance & Resistivity

Start with: $I = V_d enA$

Substitute V_d: $I = \frac{eE\tau}{m} enA = \frac{e^2 EnA\tau}{m}$

Substitute $E = V/\ell$: $I = \frac{e^2 \left(\frac{V}{\ell}\right) nA\tau}{m} = \frac{e^2 nA\tau V}{m\ell}$

Rearrange for V/I: $\frac{V}{I} = \frac{m\ell}{e^2 nA\tau}$

$$\boxed{\therefore R = \frac{m\ell}{e^2 nA\tau}}$$

As, $R = \rho\left(\frac{\ell}{A}\right)$

Equating the two: $\frac{m\cancel{\ell}}{e^2 n\cancel{A}\tau} = \rho\left(\frac{\cancel{\ell}}{\cancel{A}}\right)$

$$\boxed{\rho = \frac{m}{e^2 n\tau}}$$

$\Rightarrow \rho \propto \frac{1}{n}$ & $\rho \propto \frac{1}{\tau}$

Vector Form of Ohm's Law & Mobility

For an e^-, $q = -e$, $\vec{V}_d = -\frac{e\vec{E}\tau}{m}$

$$\vec{J} = \vec{V}_d \cdot q \cdot n = \left(\frac{-e\vec{E}\tau}{m}\right)(-e)n = \frac{e^2 E \tau n}{m}$$

Also, $\rho = \frac{m}{e^2 \tau n}$

$$\therefore \vec{J} = \frac{\vec{E}}{\rho} = \sigma\vec{E}$$

This is vector form of ohm's law.

Mobility (μ) of charge carriers:

Conductivity of any material is due to its mobile charge carriers. These can be e^- in metals, +ve & -ve in electrolyte. In semiconductors, there are e^- & holes.

Mobility is drift velocity per unit electric field ($\vec{E}$):

$$\mu = \frac{V_d}{E}$$

Also, $V_d = \frac{E e \tau}{m}$ $\quad \therefore \mu = \frac{e\tau}{m}$

Current, Mobility & Temperature

For e^-, $\mu_e = \frac{e\tau_e}{m_e}$. For holes, $\mu_h = \frac{h\tau_h}{m_h}$

In Metals: $I = V_d enA$ or $I = \mu_e \cdot E \cdot e \cdot n \cdot A$ (Since $\mu = \frac{V_d}{E}$)

In Semi-conductors: $I = I_e + I_h = enA\mu_e E + epA\mu_h E$

$$\therefore I = eAE(n\mu_e + p\mu_h)$$

Also, $I = \frac{V}{R} = \frac{E\ell A}{\rho\ell} = \frac{EA}{\rho}$.

Equating gives: $\frac{\cancel{EA}}{\rho} = e\cancel{AE}(n\mu_e + p\cancel{\mu_h})$

$$\therefore \sigma = e(n\mu_e + p\cancel{\mu_h})$$

Resistivity dependence on temperature:

As we know, $\rho = \frac{m}{ne^2\tau} \Rightarrow \rho \propto \frac{1}{\tau}$

So, $\rho(\uparrow) \rightarrow \tau(\downarrow)$ or $\rho(\downarrow) \rightarrow \tau(\uparrow)$.

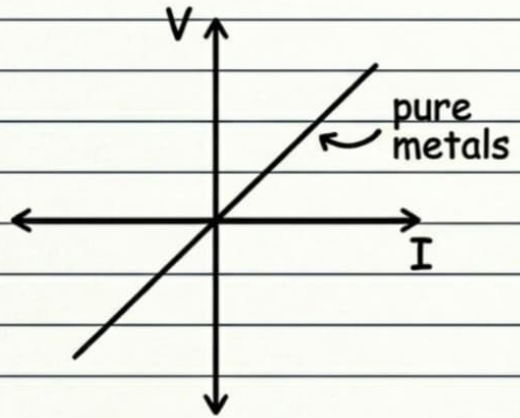
Also, if temp$(\uparrow) \rightarrow \tau(\downarrow)$. $\Rightarrow \rho(\uparrow) \rightarrow$ temp$(\uparrow)$.

$\Rightarrow \rho \propto$ temperature. Similarly, $R \propto$ temperature.

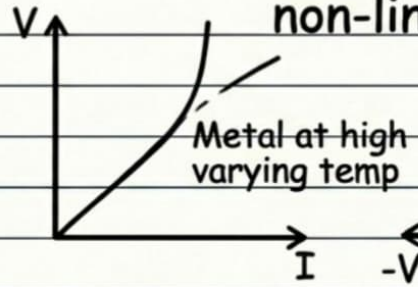
Temperature Formulas & Material Behaviors

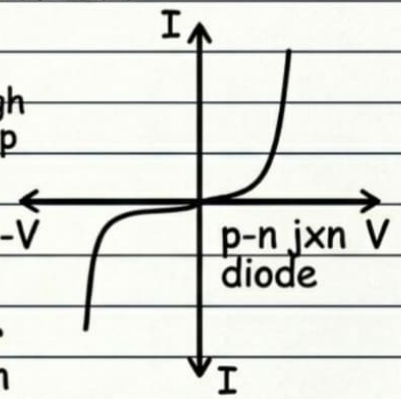
$\therefore R = R_0[1 + \alpha\Delta t]$

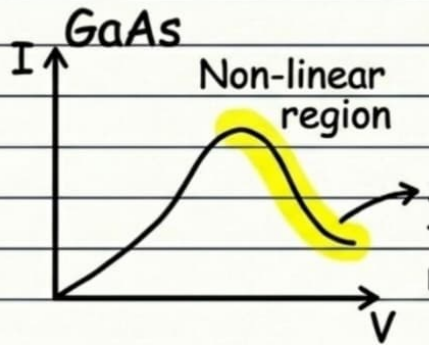
Ohmic Conductors:
Straight line graph passing through origin. $V \propto I$.



Non-Ohmic Conductors:
Non-Ohmic: Graph b/w V & I is non-linear.

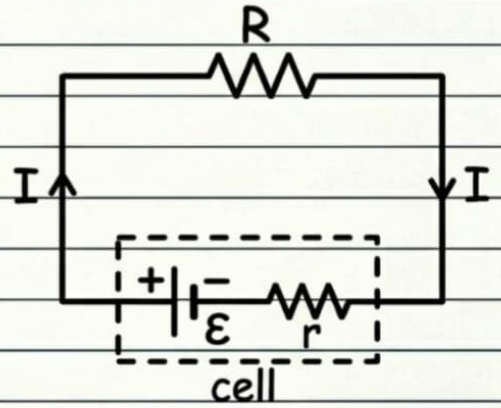






Superconductivity & Internal Cell Resistance

Superconductivity: If temp inc, resistance inc. Thus, on dec temp at very low position, resistance will reach to 0. At this low temp, conductors become Supercond & this temp is called critical temp (T_c).



Internal Resistance (r) of Cell:

1. When the charges move from low potential in cell to high potential, it experiences charges & neutral atoms in its path creating resistance for flow of positive charges.
2. Depends on:
   (i) Conc. of electrolyte.
   (ii) Distance b/w electrodes.
   (iii) Nature of electrolyte.
   (iv) Area $\propto \frac{1}{T}$ of electrolyte.

As, $\varepsilon = V + V'$ or

$\varepsilon = IR + Ir$

$\Rightarrow \varepsilon = I(R + r)$

$$\Rightarrow \boxed{I = \frac{\varepsilon}{R + r}}$$

Also, $\varepsilon - V' = V$ or

$IR = \varepsilon - Ir$

Cells in Parallel: The Derivation

Here, $V_{B_1} - V_{B_2} = V = \varepsilon_1 - I_1 r_1 \quad \Rightarrow \quad I_1 r_1 = \varepsilon_1 - V \quad \Rightarrow \quad I_1 = \dfrac{\varepsilon_1 - V}{r_1}$

Similarly, $I_2 = \dfrac{\varepsilon_2 - V}{r_2}$

$$I = I_1 + I_2$$

or $$I = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

or $$I = \frac{(\varepsilon_1 - V)r_2 + (\varepsilon_2 - V)r_1}{r_1 \cdot r_2}$$

$$I = \frac{\varepsilon_1 r_2 - V r_2 + \varepsilon_2 r_1 - V r_1}{r_1 \cdot r_2}$$

$$I = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 r_2} - V\left[\frac{r_1 + r_2}{r_1 r_2}\right]$$

or $$V\left[\frac{r_1 + r_2}{r_1 r_2}\right] = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 r_2} - I$$

or $$V = \left[\frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}\right] - I\left[\frac{r_1 \cdot r_2}{r_1 + r_2}\right]$$

or $$V = \varepsilon_{eq} - I \cdot r_{eq}$$

Identical Cells & Kirchhoff's Laws

Where, $\varepsilon_{eq} = \dfrac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$ & $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$

For n identical cells: $\varepsilon_{eq} = \varepsilon$ & $r_{eq} = \dfrac{r}{n}$ $\therefore I = \dfrac{\varepsilon}{R + \frac{r}{n}}$

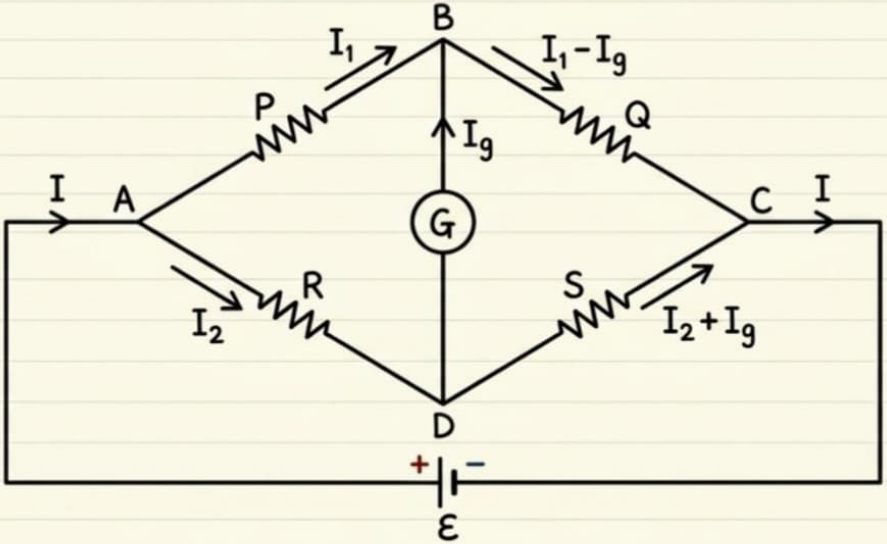
Kirchhoff's Laws:

There are 2 laws which are used in this method:

i) KCL (Kirchhoff's current law): Law of conservation of charge based. $\sum I = 0$.

ii) KVL (Kirchhoff's voltage law): Law of conservation of energy based.
$\sum V = \sum(IR) = 0$.

The Wheatstone Bridge (Setup)



Under balanced condition, there is no p.d across galvanometer, thus no current flows through it. ($I_g = 0$).

$$\boxed{\text{Under balanced condition: } \frac{P}{Q} = \frac{R}{S}}$$

The Wheatstone Proof

In loop ABDA:

$V_A - I_1P - I_gG + I_2R = V_A$

$-I_1P - I_gG + I_2R = 0$

or $-I_1P + I_2R = 0$ (since $I_gG = 0$)

or $I_1P = I_2R$ --- (Equation I)

Also, in loop BCDB:

$V_B - Q(I_1 - I_g) + S(I_2 + I_g) + I_gG = V_B$

$-Q(I_1 - I_g) + S(I_2 + I_g) = 0$ (since $I_gG = 0$)

$Q(I_1 - I_g) = S(I_2 + I_g)$

$I_1Q = I_2S$ (since $I_g = 0$) --- (Equation II)

Dividing (I) by (II):

$$\frac{I_1P}{I_1Q} = \frac{I_2R}{I_2S} \Rightarrow \boxed{\frac{P}{Q} = \frac{R}{S}}$$

In such cases $I_g = 0$.